

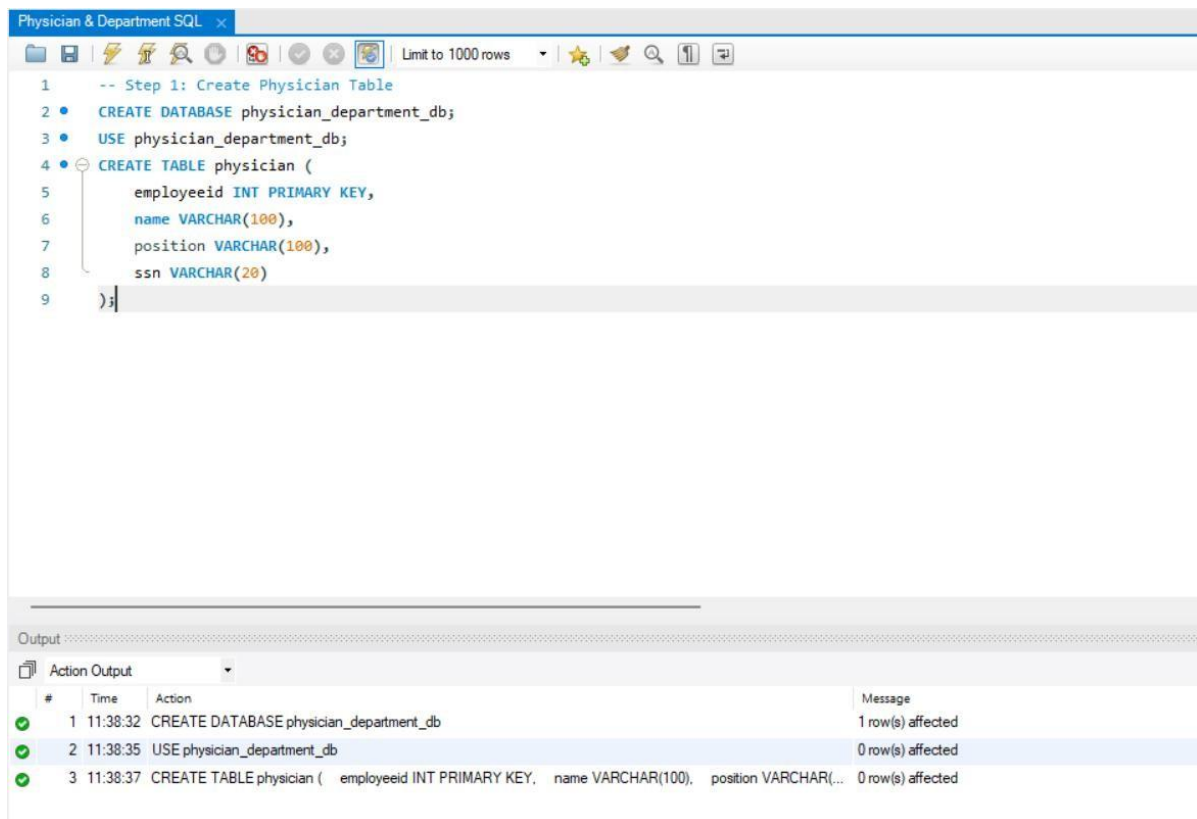
PHYSICIAN & DEPARTMENT SQL

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Question :

Perform SQL operations on Physician and Department tables using INNER JOIN, WHERE, LIKE, NOT LIKE, IN, NOT IN, EXISTS, GROUP BY, HAVING, ORDER BY, subqueries, COUNT(), COUNT(DISTINCT), MIN(), and MAX() to retrieve and analyse physician and department information.

Step 1 : Create Physician Database and Table



The screenshot shows a SQL IDE window titled "Physician & Department SQL". The SQL editor contains the following code:

```
1  -- Step 1: Create Physician Table
2  • CREATE DATABASE physician_department_db;
3  • USE physician_department_db;
4  • CREATE TABLE physician (
5      employeeid INT PRIMARY KEY,
6      name VARCHAR(100),
7      position VARCHAR(100),
8      ssn VARCHAR(20)
9  );
```

Below the editor is the "Output" pane, which shows the execution results:

#	Time	Action	Message
✓ 1	11:38:32	CREATE DATABASE physician_department_db	1 row(s) affected
✓ 2	11:38:35	USE physician_department_db	0 row(s) affected
✓ 3	11:38:37	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected

JoinsSQL Aggregate functions sql SQL File 4*

Limit to 1000 rows

```

3 • USE physician_department_db;
4 • CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10 -- Step 2: Insert Data into Physician
11 • INSERT INTO physician VALUES
12     (1,'Angela','Staff Internist','111111111'),
13     (2,'Smija','Attending Physician','222222222'),
14     (3,'Joel','Surgical Attending Physician','333333333'),
15     (4,'Andrea','Senior Attending Physician','444444444'),
16     (5,'Advika','Head Chief of Medicine','555555555'),
17     (6,'Tianne','Surgical Attending Physician','666666666'),
18     (7,'Evelyn','Surgical Attending Physician','777777777'),
19     (8,'Elizebth','MD Resident','888888888'),
20     (9,'Riya','Attending Psychiatrist','999999999');
21 -- Step 3: Create Department Table
22 • CREATE TABLE department (
23     departmentid INT PRIMARY KEY,

```

Output

Action Output

#	Time	Action	Message
✓ 1	22:13:48	USE physician_department_db	0 row(s) affected
✓ 2	22:13:52	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100), ssn VARCHAR(20))	0 row(s) affected
✓ 3	22:14:02	INSERT INTO physician VALUES (1,'Angela','Staff Internist','111111111'), (2,'Smija','Attending Physician','222222222'), (3,'Joel','Surgical Attending Physician','333333333'), (4,'Andrea','Senior Attending Physician','444444444'), (5,'Advika','Head Chief of Medicine','555555555'), (6,'Tianne','Surgical Attending Physician','666666666'), (7,'Evelyn','Surgical Attending Physician','777777777'), (8,'Elizebth','MD Resident','888888888'), (9,'Riya','Attending Psychiatrist','999999999');	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0

JoinsSQL Aggregate functions sql SQL File 4*

Limit to 1000 rows

```

9 );
10 -- Step 2: Insert Data into Physician
11 • INSERT INTO physician VALUES
12     (1,'Angela','Staff Internist','111111111'),
13     (2,'Smija','Attending Physician','222222222'),
14     (3,'Joel','Surgical Attending Physician','333333333'),
15     (4,'Andrea','Senior Attending Physician','444444444'),
16     (5,'Advika','Head Chief of Medicine','555555555'),
17     (6,'Tianne','Surgical Attending Physician','666666666'),
18     (7,'Evelyn','Surgical Attending Physician','777777777'),
19     (8,'Elizebth','MD Resident','888888888'),
20     (9,'Riya','Attending Psychiatrist','999999999');
21 -- Step 3: Create Department Table
22 • CREATE TABLE department (
23     departmentid INT PRIMARY KEY,
24     name VARCHAR(100),
25     head INT
26 );
27 -- Step 4: Insert Data into Department
28 • INSERT INTO department VALUES
29     (1,'General Medicine',4);

```

Output

Action Output

#	Time	Action	Message
✓ 1	22:13:48	USE physician_department_db	0 row(s) affected
✓ 2	22:13:52	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100), ssn VARCHAR(20))	0 row(s) affected
✓ 3	22:14:02	INSERT INTO physician VALUES (1,'Angela','Staff Internist','111111111'), (2,'Smija','Attending Physician','222222222'), (3,'Joel','Surgical Attending Physician','333333333'), (4,'Andrea','Senior Attending Physician','444444444'), (5,'Advika','Head Chief of Medicine','555555555'), (6,'Tianne','Surgical Attending Physician','666666666'), (7,'Evelyn','Surgical Attending Physician','777777777'), (8,'Elizebth','MD Resident','888888888'), (9,'Riya','Attending Psychiatrist','999999999');	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓ 4	22:14:32	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected

JoinsSQL Aggregate functions sql SQL File 4*

```

12 (1,'Angela','Staff Internist','11111111'),
13 (2,'Smija','Attending Physician','22222222'),
14 (3,'Joel','Surgical Attending Physician','33333333'),
15 (4,'Andrea','Senior Attending Physician','44444444'),
16 (5,'Advika','Head Chief of Medicine','55555555'),
17 (6,'Tianne','Surgical Attending Physician','66666666'),
18 (7,'Evelyn','Surgical Attending Physician','77777777'),
19 (8,'Elizebth','MD Resident','88888888'),
20 (9,'Riya','Attending Psychiatrist','99999999');
21 -- Step 3: Create Department Table
22 CREATE TABLE department (
23     departmentid INT PRIMARY KEY,
24     name VARCHAR(100),
25     head INT
26 );
27 -- Step 4: Insert Data into Department
28 INSERT INTO department VALUES
29 (1,'General Medicine',4),
30 (2,'Surgery',7),
31 (3,'Psychiatry',9);
32 -- Step 5: INNER JOIN

```

Output

#	Time	Action	Message
1	22:13:48	USE physician_department_db	0 row(s) affected
2	22:13:52	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
3	22:14:02	INSERT INTO physician VALUES (1,'Angela','Staff Internist','11111111'), (2,'Smija','Attending Physician','22222...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
4	22:14:32	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
5	22:15:00	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Step 2 : JOIN

JoinsSQL Aggregate functions sql SQL File 4*

```

29 (1,'General Medicine',4),
30 (2,'Surgery',7),
31 (3,'Psychiatry',9);
32 -- Step 5: INNER JOIN
33 SELECT d.name AS Department,
34        p.name AS Head_Physician
35 FROM department d
36 JOIN physician p
37 ON d.head = p.employeeid;

```

Result Grid

Department	Head_Physician
General Medicine	Andrea
Surgery	Evelyn
Psychiatry	Riya

Result 1 x

Output

#	Time	Action	Message
1	22:13:48	USE physician_department_db	0 row(s) affected
2	22:13:52	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
3	22:14:02	INSERT INTO physician VALUES (1,'Angela','Staff Internist','11111111'), (2,'Smija','Attending Physician','22222...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
4	22:14:32	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
5	22:15:00	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6	22:15:37	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d....	3 row(s) returned

Step 3 : WHERE

JoinsSQL Aggregate functions sql SQL File 4*

```
35 FROM department d
36 JOIN physician p
37 ON d.head = p.employeeid;
38 -- Step 6: WHERE Clause
39 • SELECT *
40 FROM physician
41 WHERE position = 'Surgical Attending Physician';
42 -- Step 7: WHERE + LIKE
43 • SELECT *
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	employeeid	name	position	ssn
▶	3	Joel	Surgical Attending Physician	333333333
	6	Tianne	Surgical Attending Physician	666666666
	7	Evelyn	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

physician 2 ×

Output

Action Output

#	Time	Action	Message
2	22:13:52	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100))	0 row(s) affected
3	22:14:02	INSERT INTO physician VALUES (1,'Angela','Staff Internist','111111111'), (2,'Smija','Attending Physician','222222222'), (3,'John','Attending Physician','333333333')	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
4	22:14:32	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
5	22:15:00	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6	22:15:37	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid	3 row(s) returned
7	22:16:03	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned

Step 4 : LIKE

Physician & Department SQL

```
37 ON d.head = p.employeeid;
38 -- Step 6: WHERE Clause
39 • SELECT *
40 FROM physician
41 WHERE position = 'Surgical Attending Physician';
42 -- Step 7: WHERE + LIKE
43 • SELECT *
44 FROM physician
45 WHERE name LIKE 'John%';
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	employeeid	name	position	ssn
*	NULL	NULL	NULL	NULL

physician 3 ×

Output

Action Output

#	Time	Action	Message
4	11:41:45	INSERT INTO physician VALUES (1,'Harshika','Staff Internist','111111111'), (2,'Ravinder','Attending Physician','222222222'), (3,'John','Attending Physician','333333333')	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
5	11:42:13	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
6	11:43:08	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7	11:44:23	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid	3 row(s) returned
8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned

Step 5 : COUNT

Physician & Department SQL

```
40 FROM physician
41 WHERE position = 'Surgical Attending Physician';
42 -- Step 7: WHERE + LIKE
43 • SELECT *
44 FROM physician
45 WHERE name LIKE 'John%';
46 -- Step 8: COUNT()
47 • SELECT COUNT(*) AS Total_Physicians
48 FROM physician;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: I

Total_Physicians
9

Result 4

Output

Action Output

#	Time	Action	Message
✓ 5	11:42:13	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓ 6	11:43:08	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	11:44:23	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
✓ 10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned

Physician & Department SQL

```
43 • SELECT *
44 FROM physician
45 WHERE name LIKE 'John%';
46 -- Step 8: COUNT()
47 • SELECT COUNT(*) AS Total_Physicians
48 FROM physician;
49 -- Step 9: COUNT DISTINCT
50 • SELECT COUNT(DISTINCT position) AS Different_Positions
51 FROM physician;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: I

Different_Positions
7

Result 5

Output

Action Output

#	Time	Action	Message
✓ 6	11:43:08	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	11:44:23	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓ 9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
✓ 10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned

Step 6 : GROUP BY

Physician & Department SQL

```
48 FROM physician;
49 -- Step 9: COUNT DISTINCT
50 • SELECT COUNT(DISTINCT position) AS Different_Positions
51 FROM physician;
52 -- Step 10: GROUP BY
53 • SELECT position,
54       COUNT(*) AS Total_Employees
55 FROM physician
56 GROUP BY position;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 6

Output

Action Output

#	Time	Action	Message
7	11:44:23	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned

Physician & Department SQL

```
54       COUNT(*) AS Total_Employees
55 FROM physician
56 GROUP BY position;
57 -- Step 11: GROUP BY + HAVING
58 • SELECT position,
59       COUNT(*) AS Total_Employees
60 FROM physician
61 GROUP BY position
62 HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Surgical Attending Physician	3

Result 7

Output

Action Output

#	Time	Action	Message
8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

Step 7: ORDER BY

JoinsSQL Aggregate functions sql SQL File 4* x

Limit to 1000 rows

```
61 GROUP BY position
62 HAVING COUNT(*) > 1;
63 -- Step 12: ORDER BY ASC
64 • SELECT *
65 FROM physician
66 ORDER BY name ASC;
67 -- Step 13: ORDER BY DESC
68 • SELECT *
69 FROM physician
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	employeeid	name	position	ssn
▶	5	Advika	Head Chief of Medicine	555555555
	4	Andrea	Senior Attending Physician	444444444
	1	Angela	Staff Internist	111111111
	8	Elizabeth	MD Resident	888888888
	7	Evelyn	Surgical Attending Physician	777777777
	3	Joel	Surgical Attending Physician	333333333
	9	Riya	Attending Psychiatrist	999999999
	2	Smija	Attending Physician	222222222
	6	Tianne	Surgical Attending Physician	666666666
•	NULL	NULL	NULL	NULL

physician 8 x

Output

Action Output

#	Time	Action	Message
✓	8 22:16:54	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
✓	9 22:16:59	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓	10 22:17:02	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓	11 22:17:07	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓	12 22:17:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓	13 22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned

Step 7: ORDER BY

JoinsSQL Aggregate functions sql SQL File 4*

Limit to 1000 rows

```
63 -- Step 12: ORDER BY ASC
64 • SELECT *
65 FROM physician
66 ORDER BY name ASC;
67 -- Step 13: ORDER BY DESC
68 • SELECT *
69 FROM physician
70 ORDER BY employeeid DESC;
71 -- Step 14: WHERE + IN
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: [IA](#)

	employeeid	name	position	ssn
▶	9	Riya	Attending Psychiatrist	999999999
	8	Elizebth	MD Resident	888888888
	7	Evelyn	Surgical Attending Physician	777777777
	6	Tianne	Surgical Attending Physician	666666666
	5	Advika	Head Chief of Medicine	555555555
	4	Andrea	Senior Attending Physician	444444444
	3	Joel	Surgical Attending Physician	333333333
	2	Smija	Attending Physician	222222222
	1	Angela	Staff Internist	111111111
*	NULL	NULL	NULL	NULL

physician 9 x

Output

Action Output

#	Time	Action	Message
✓ 9	22:16:59	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 10	22:17:02	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	22:17:07	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 12	22:17:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned

Step 8 : IN

```
70 ORDER BY employeeid DESC;
71 -- Step 14: WHERE + IN
72 • SELECT *
73 FROM physician
74 WHERE employeeid IN
75 (
76     SELECT head
77     FROM department
78 );
```

Result Grid

	employeeid	name	position	ssn
▶	4	Andrea	Senior Attending Physician	444444444
	7	Evelyn	Surgical Attending Physician	777777777
	9	Riya	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 10 x

Output

Action Output

#	Time	Action	Message
✓ 10	22:17:02	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	22:17:07	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 12	22:17:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned

Step 9 : NOT IN

Step 8 : IN

JoinsSQL Aggregate functions sql SQL File 4* x

Limit to 1000 rows

```
78 );
79 -- Step 15: NOT IN
80 • SELECT *
81 FROM physician
82 WHERE employeeid NOT IN
83 (
84     SELECT head
85     FROM department
86 );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employeeid	name	position	ssn
▶	1	Angela	Staff Internist	111111111
	2	Smija	Attending Physician	222222222
	3	Joel	Surgical Attending Physician	333333333
	5	Advika	Head Chief of Medicine	555555555
	6	Tianne	Surgical Attending Physician	666666666
	8	Elizbeth	MD Resident	888888888
*	NULL	NULL	NULL	NULL

physician 11 x

Output

Action Output

#	Time	Action	Message
✓ 11	22:17:07	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 12	22:17:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned

Step 10 : GROUP BY + ORDER BY

Physician & Department SQL

```
84 SELECT head
85 FROM department
86 );
87 -- Step 16: GROUP BY + ORDER BY
88 • SELECT position,
89     COUNT(*) AS Total_Employees
90 FROM physician
91 GROUP BY position
92 ORDER BY Total_Employees DESC;
```

Result Grid

	position	Total_Employees
▶	Surgical Attending Physician	3
	Staff Internist	1
	Attending Physician	1
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

Result 12

Output

Action Output

#	Time	Action	Message
✓ 13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 14	11:48:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 15	11:49:03	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 16	11:53:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 17	11:53:34	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 18	11:54:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned

Step 11: LIKE

Step 10 : GROUP BY + ORDER BY

```
JoinsSQL  Aggregate functions sql  SQL File 4* x
Limit to 1000 rows

88 • SELECT position,
89     COUNT(*) AS Total_Employees
90 FROM physician
91 GROUP BY position
92 ORDER BY Total_Employees DESC;
93 -- Step 17: WHERE + LIKE
94 • SELECT *
95 FROM physician
96 WHERE position LIKE '%Attending%';
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
employeeid	name	position	ssn	
2	Smija	Attending Physician	222222222	
3	Joel	Surgical Attending Physician	333333333	
4	Andrea	Senior Attending Physician	444444444	
6	Tianne	Surgical Attending Physician	666666666	
7	Evelyn	Surgical Attending Physician	777777777	
9	Riya	Attending Psychiatrist	999999999	
*	NULL	NULL	NULL	

#	Time	Action	Message
✓ 13	22:17:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:19:09	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned

Step 12: JOIN + WHERE

JoinsSQL Aggregate functions sql SQL File 4* x

Limit to 1000 rows

```
94 • SELECT *
95 FROM physician
96 WHERE position LIKE '%Attending%';
97 -- Step 18: JOIN + WHERE
98 • SELECT p.name
99 FROM physician p
100 JOIN department d
101 ON p.employeeid = d.head
102 WHERE d.name = 'Surgery';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [F1](#)

name
Evelyn

Result 14 x

Output

Action Output

#	Time	Action	Message
✓ 14	22:17:49	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:19:09	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned
✓ 19	22:19:30	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned

Step 13: JOIN+ORDER BY

JoinsSQL Aggregate functions sql SQL File 4* x

Limit to 1000 rows

```
101 ON p.employeeid = d.head
102 WHERE d.name = 'Surgery';
103 -- Step 19: JOIN + ORDER BY
104 • SELECT d.name AS Department,
105        p.name AS Head_Physician
106 FROM department d
107 JOIN physician p
108 ON d.head = p.employeeid
109 ORDER BY d.name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [F1](#)

Department	Head_Physician
General Medicine	Andrea
Psychiatry	Riya
Surgery	Evelyn

Result 15 x

Output

Action Output

#	Time	Action	Message
✓ 15	22:18:15	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:19:09	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned
✓ 19	22:19:30	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 20	22:19:56	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned

Step 14: NOT LIKE

JoinsSQL Aggregate functions sql SQL File 4* x

Limit to 1000 rows

```
106 FROM department d
107 JOIN physician p
108 ON d.head = p.employeeid
109 ORDER BY d.name;
110 -- Step 20: WHERE + NOT LIKE
111 • SELECT *
112 FROM physician
113 WHERE position NOT LIKE '%Surgical%';
114 -- Step 21: COUNT + HAVING
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	employeeid	name	position	ssn
▶	1	Angela	Staff Internist	111111111
	2	Smija	Attending Physician	222222222
	4	Andrea	Senior Attending Physician	444444444
	5	Advika	Head Chief of Medicine	555555555
	8	Elizbeth	MD Resident	888888888
	9	Riya	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 16 x

Output

Action Output

#	Time	Action	Message
✓ 16	22:18:38	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:19:05	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:19:09	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned
✓ 19	22:19:30	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 20	22:19:56	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 21	22:20:34	SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000	6 row(s) returned

Step 15: COUNT+HAVING

Physician & Department SQL x

Limit to 1000 rows

```
111 • SELECT *
112 FROM physician
113 WHERE position NOT LIKE '%Surgical%';
114 -- Step 21: COUNT + HAVING
115 • SELECT position,
116 COUNT(*) AS Total
117 FROM physician
118 GROUP BY position
119 HAVING COUNT(*) >= 2;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	position	Total
▶	Surgical Attending Physician	3

Result 17 x

Output

Action Output

#	Time	Action	Message
✓ 18	11:54:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 19	11:54:38	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned
✓ 20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000	6 row(s) returned
✓ 23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned

Step 16: EXISTS

Physician & Department SQL

```
120  -- Step 22: EXISTS
121  SELECT *
122  FROM department d
123  WHERE EXISTS
124  (
125      SELECT 1
126      FROM physician p
127      WHERE p.employeeid = d.head
128  );
```

Result Grid

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

department 18

Output

Action Output

#	Time	Action	Message
19	11:54:38	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned
20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000	6 row(s) returned
23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
24	12:00:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employee...	3 row(s) returned

Step 17: MAX

Physician & Department SQL

```
123  WHERE EXISTS
124  (
125      SELECT 1
126      FROM physician p
127      WHERE p.employeeid = d.head
128  );
129  -- Step 23: MAX()
130  SELECT MAX(employeeid) AS Highest_Employee_ID
131  FROM physician;
```

Result Grid

Highest_Employee_ID
9

Result 19

Output

Action Output

#	Time	Action	Message
20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000	6 row(s) returned
23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
24	12:00:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employee...	3 row(s) returned
25	12:00:41	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned

Step 18: MIN

Physician & Department SQL

```
126     FROM physician p
127     WHERE p.employeeid = d.head
128 );
129 -- Step 23: MAX()
130 • SELECT MAX(employeeid) AS Highest_Employee_ID
131     FROM physician;
132 -- Step 24: MIN()
133 • SELECT MIN(employeeid) AS Lowest_Employee_ID
134     FROM physician;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: IA

Lowest_Employee_ID
1

Result 20 x

Output

Action Output

#	Time	Action	Message
✓ 21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓ 23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
✓ 24	12:00:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned
✓ 25	12:00:41	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 26	12:01:06	SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned